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Research Article

Ayurvedic Herbal Patch

Development and Preliminary Clinical Assessment of an Ayurvedic Herbal Patch for Menstrual Pain Management

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Introduction: Uterine cramping during menstruation affects a vast majority of reproductive-age women without pelvic abnormalities, drastically disrupting their regular daily routines. Standard clinical options rely heavily on oral non-steroidal anti-inflammatory drugs (NSAIDs) like naproxen sodium, yet these medications frequently trigger gastrointestinal distress, exhibit delayed systemic absorption, and suffer from poor patient compliance. Seeking a superior therapeutic route, this investigation focused on engineering and testing an innovative polyherbal transdermal drug delivery system. This localized patch serves as an inherently safer, non-invasive, and highly comfortable alternative designed to optimize real-world menstrual pain relief.

Materials and Methods: Prepared patches were fabricated by the solvent- casting method, incorporating five Ayurvedic herbal actives, namely, Peppermint Satva (Mentha piperita), Shallaki Guggul (Boswellia serrata), Taila Parni (Eucalyptus globulus), Haridra Satva (Curcuma longa), and Cinnamon oil (Cinnamomum verum). Physical and mechanical characterization encompassed folding endurance, thickness, weight uniformity, and percentage elongation. Transdermal permeation behaviour was investigated using a Franz diffusion cell assembly with porcine skin.

Results: The prepared patches demonstrated favourable results in the areas of their physicochemical properties, mechanical flexibility and their distribution was much consistent. The folding endurance of the patches averaged (100.33 ± 1.52), thickness (0.63 ± 0.90 mm), weight variation (3.31 ± 0.20 g), and percentage elongation (45%) indicated that they had retained their structural integrity. The data from the ex-vivo permeation studies showed that the patches had a biphasic release pattern with a total duration of 12 hours and a total of 91.23% cumulative drug release. The overall substance that permeated through each patch over time was 13687 µg/patch whilst the maximum steadystate flux was found to be 465.51 µg/cm²/hr. The clinical study showed that the in vitro data was confirmed since 96% of the participants noted they had experienced pain relief from their menstrual symptoms within 30 minutes of application.

Conclusion: Collectively, these results support the developed Ayurvedic polyherbal transdermal patch as an effective, non- invasive alternative for menstrual pain management, offering sustained active delivery, rapid therapeutic onset, and a reduced systemic side effect burden compared to conventional oral analgesics.

Keywords: Ayurvedic herbal patch, Menstrual pain, Menstrual patch, Transdermal drug delivery system, Transdermal Patches

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Introduction

Menstrual pain typically manifests as lower abdominal cramping that sets in just before or at the onset of menstruation, usually subsiding within a few days. When severe, these cramps deeply disrupt standard daily routines and severely compromise an individual's physical and emotional well-being.**[1]** Menstruation is a normal physiological process that begins at menarche, generally around the age of 11 years, and continues until menopause. During the menstrual cycle, many women experience menstrual pain associated with abdominal cramps, nausea, fatigue, a headache, backache, and emotional discomfort.**[2,3]** Menstrual pain occurs in the absence of any pelvic pathology and typically lasts from eight to 72 hours.**[4]**It mainly occurs due to increased prostaglandin F2α production, which stimulates uterine contractions and vasoconstriction, resulting in reduced uterine blood flow, hypoxia, and pain.**[5]**

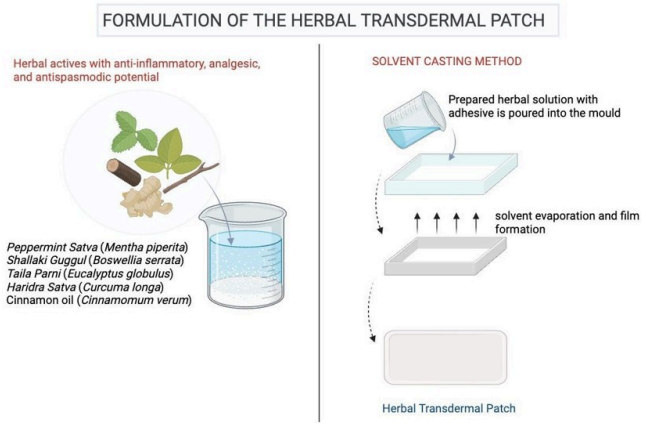


Fig. 1: Graphical abstract illustrating the formulation and mechanism of action of the herbal transdermal patch for menstrual pain management. The image was made using the BioRender illustration tool (*BioRender, Toronto, ON, Canada*).

The occurrence of menstrual pain among adolescents and young women is reported to be as high as 50-70% worldwide.**[6]** Conventional management commonly involves oral administration of NSAIDs such as Naproxen Sodium, and oral contraceptives (OCs).**[7]** However, long term use through oral means is associated with side effects e.g., irritation of Gastrointestinal (GI) tract, gallbladders, and/or kidneys, delayed onset, and unable to dose frequently; therefore, patients do not comply with these regimens.**[8]**

As a result, an ideal management for menstrual pain would provide rapid pain relief while minimizing systemic side effects and improving patient adherence to the therapeutic regimen.

Transdermal drug delivery systems (TDDS) have become a viable alternative to oral medications over recent years and are being used to treat many conditions, including cardiovascular and central nervous system disorders, hormonal therapy, and pain management.**[9-11]** The use of transdermal patches as non-invasive delivery methods that do not use the gastrointestinal tract means that there are fewer GI-related side effects since the drug is delivered directly into the bloodstream via the skin. The direct delivery of a drug to the bloodstream via a patch allows for controlled and sustained release of the drug into the body, resulting in better therapeutic outcomes and greater adherence to management.**[12,13]** Additionally, herbal medicine is gaining more popularity because it is derived from nature, has been clinically proven to work, and is regarded as safe. Compared to traditional medications, herbal formulations typically produce fewer adverse effects and are therefore better suited to treating menstrual pain on an ongoing basis.**[14]**This has led to growing interest in the development of herbal-based transdermal patches as safer and more patient-friendly alternatives to conventional medications.

In the present study, an herbal transdermal patch containing *Peppermint Satva*, *Shallaki Guggul*, *Taila Parni*, *Haridra Satva*, and Cinnamon Oil was developed for the management of menstrual pain. Due to the menthol and menthone present in it, *Peppermint Satva* (*Mentha piperita*) has analgesic (pain-relieving), antispasmodic (muscle-relaxing), and anti-inflammatory effects, which can help relieve uterine smooth muscle cramping and pain. Menthol additionally activates cold-sensitive receptors (TRPM8), creating a cooling sensation, which affects how we perceive pain. As a result, peppermint may provide fast relief from pain through a non-opioid pathway without causing central nervous system depression.**[15,16]***Shallaki Guggul* (*Boswellia serrata*) is an effective anti-inflammatory herb with much anti-inflammatory action via Boswellic acid (AKBA), which decreases leukotriene synthesis through inhibition of the 5-lipoxygenase pathway without the stomach upset associated with non-steroidal anti-inflammatory drugs.**[17]**

Taila Parni (*Eucalyptus globulus*) enhances skin penetration of various constituents by modifying lipid structure in the stratum corneum, allowing the skin to absorb more of the active ingredients of the various topical formulations.**[18]** In addition, Eucalyptol has anti-inflammatory and mild anti-microbial properties, supporting the efficacy of the formulation.**[19,20]**

Haridra Satva derived from *Curcuma longa*, is a well-recognised anti-inflammatory agent that inhibits NF-κB, downregulates COX-2, and help in suppressing pro-inflammatory cytokines such as TNF-α, IL-6, and IL-1β, thereby helping manage inflammatory menstrual pain.**[21]** Curcumin also works as a powerful antioxidant. As a shield against oxidative damage, it supports the body's production of natural antioxidants and clears away destructive reactive oxygen species (ROS).**[22]** Cinnamaldehyde is one of the active constituents in cinnamon oil; it has antispasmodic and anti-inflammatory properties through its ability to relax smooth uterine muscles, decrease prostaglandin synthesis, and inhibit the production of inflammatory cytokines. In this way, cinnamon oil can help reduce menstrual pain.**[23]**

The aim of this study was to create an alternative method for obtaining menstrual pain relief via improved methods, through the application of innovative technologies, in the form of a transdermal patch containing herbal ingredients. These active components assist with the alleviation of menstrual discomfort and improve upon typical methods for managing this type of pain by providing a higher efficacy rate, greater patient adherence, and lower side effects. To bridge the gap between in-vitro performance and real-world compliance, the laboratory findings of the optimized formulation were systematically correlated with an in-vivo user acceptability assessment in cohort of menstruating individuals experiencing menstrual pain.

Aim and Objectives

The aim was to develop and evaluate an herbal transdermal patch for the management of menstrual pain and check its potential as a non-invasive alternative to conventional oral medications.

1. Prepare an herbal transdermal patch containing *Peppermint Satva* (*Mentha piperita*), *Shallaki Guggul* (*Boswellia serrata*), *Taila Parni* (*Eucalyptus globulus*), *Haridra Satva* (*Curcuma longa*), And Cinnamon Oil, & evaluate its physicochemical, mechanical, and ex-vivo permeation characteristics.

2. To check the stability of the optimized herbal transdermal patch under accelerated and long-term storage conditions.

3. To evaluate the preliminary user acceptability, pain relief response, skin tolerability, and adhesion performance of the developed patch in individuals experiencing menstrual pain.

Materials and Methods

Peppermint Satva (*Mentha piperita* extract) was obtained from Crystal Pharma, *Shallaki Guggul* (*Boswellia serrata* resin) was obtained from Bangalore Sales Corporation, *Taila Parni* (*Eucalyptus globulus* oil) was purchased from Nature Natural India and Falcon Pharma Pvt. Ltd., *Haridra Satva* (curcumin extract from *Curcuma longa*) was obtained from Star Bio Pharma, Bangalore, and Cinnamon Oil (*Cinnamomum verum* essential oil) was obtained from Nature Natural India. Vitamin E was sourced from Crystal Pharma, Mumbai, Maharashtra, while Burium Adhesive and Aerosil were obtained from Burium Chemicals Co. Ltd. and M.S. Mundra Enterprises, respectively.

Preparation of herbal patches

Herbal transdermal patches were manufactured using a solvent-casting approach under controlled environmental conditions (temperature: 2025°C, relative humidity: 40-60%) in a classified manufacturing area. The adhesive component was formulated by dissolving 4.5 g of adhesive in five millilitres of ethyl acetate, stirring continuously on a magnetic stirrer until a clear solution formed. Polyethylene Glycol was included directly into this adhesive phase as a thin film-forming plasticizer, along with Dibutyl Phthalate to enhance dermal penetration. The five herbal actives were then sequentially incorporated into the adhesive base at the proportions detailed in Table no. 1. The complete mixture was thoroughly blended under ongoing agitation to achieve a homogeneous casting solution.

The combined casting solution was then poured onto Petri dishes or bangle-shaped moulds. Moulds were left undisturbed at room temperature for a full 24-hour period to dry, allowing solvent evaporation and a coherent film formation.

Resulting patches were detached from the mould surface and transferred to a desiccator for storage before further testing. The illustration for this process was designed by the BioRender software, which is a scientific illustration tool (*BioRender, Toronto, ON, Canada*).

Table 1: Ingredients and quantity of the Herbs used to prepare herbal patch

Ingredients	F1	F2	F3	F4	F5
Peppermint Satva (Mentha piperita)	15%	10%	12%	10%	8%
Shallaki Guggul (Boswellia serrata)	3%	4%	3%	2%	6%
Taila Parni (Eucalyptus globulus)	5%	9%	5%	6%	5%
Haridra Satva (Curcuma longa)	1%	2%	3%	3%	4%
Cinnamon oil	QS	QS	QS	QS	QS

"F" stands for formulation, whereas "QS" stands for Quantity Sufficient.

Evaluation of herbal patches

The physical and mechanical performance of the prepared patches was assessed through folding endurance, thickness measurement, weight variation, and percentage elongation.

For folding endurance, the rectangular patches (four cm * two cm) were repeatedly folded at the same location and the number of folds tolerated before fracture was recorded. The thickness of the patch was determined at randomly selected points using a calibrated digital vernier calliper, and mean values with the standard deviations were reported. Weight uniformity was assessed by cutting circular patches of 3.14 cm diameter and recording individual weights on a digital analytical balance. Percentage elongation was calculated instrumentally using an elongation testing apparatus, as done in equation 1.

$$\text{Percentage elongation} = \frac{\text{Final length of the patch} - \text{Initial length of the patch}}{\text{Initial length of the patch}} x\ 100$$

Equation 1: Equation used for calculation of percentage elongation of the transdermal patch.

Ex-vivo skin permeation studies

A Franz diffusion cell system served as the platform for assessing the transdermal permeation behaviour, with freshly excised porcine skin that was shaved prior to use, functioning as biological membrane. Skin specimens of two centimetres square were secured across the donor compartment, which had an effective diffusion surface of 0.6364 cm^2. The prepared optimised patch was positioned directly against the epidermal surface.

Phosphate-buffered saline at pH 7.4 was chosen as receptor medium, with a volume of five millilitres, thermostated at 37 degrees Celsius under constant agitation to approximate in-vivo physiological conditions. Sampling was carried out by withdrawing one milliliter aliquots from receptor phase at zero, three, six, nine, and twelve hours post- application. Each withdrawn volume was immediately refilled with fresh PBS to preserve sink conditions throughout the experiment. Drug content in each collected sample was quantified by spectrophotometric analysis. The graphs represented for this analysis were generated by the Origin software (*Originlab Corporation, Northampton, MA, USA*).

Stability Studies

Stability testing of the optimised herbal patch was performed under conditions prescribed by ICH Q1A (R2). Two independent storage arms were established, an Accelerated Stability Study where the samples were kept at 40°C ± 2°C and 75% RH ± 5% for three months and a Long- Term Stability Study where the samples were stored at 30°C ± 2°C and 65% RH ± 5% for three months. Individual patch samples were sealed within aluminium zip-lock pouches and housed in a validated stability chamber operating at the specified environmental parameters. To align with both WHO pharmaceutical product stability requirements & CDSCO regulatory guidance for Indian market, baseline characterization was performed at T0, with follow- up assessment conducted at three-month mark (T3M).

Statistical Analysis

All results were reported as values from replicate analyses. Percentage assay values were calculated relative to the labelled amount. We evaluated the stability data between the baseline and the 3-month mark using ICH Q1E standards. Since there were no major shifts in drug content or release behaviour under accelerated conditions, the patch successfully met the regulatory benchmarks for stability.

Preliminary User Acceptability Assessment

A preliminary user acceptability study was conducted with Twenty-Seven menstruating female participants (aged 14-36 years) experiencing menstrual pain. Baseline pain was documented prior to application, with the majority categorizing their pain as severe and identifying the lower abdomen as the primary target site.

Participants applied the Ayurvedic transdermal patch topically for an intended wear duration of four to 12 hours under normal daily activity conditions. Patient-reported outcomes (PROs) and acceptability parameters were evaluated via standardized questionnaires across the multiple metrics. Analgesic efficacy was evaluated by recording the onset of noticeable pain relief (within 15 min, 15–30 min, and >30 min). Perceived global efficacy was quantified using a five-point Likert scale (one = no relief; five = complete pain elimination). Subjective physical sensations (e.g., cooling effects) and the therapeutic perception of the herbal scent were documented to evaluate sensory & organoleptic profiles. The adhesive performance was checked by analysing structural integrity and secure attachment throughout the active wear window. Local dermal safety was monitored during wear and post-removal to evaluate potential adverse effects, including subcutaneous irritation or transient erythema to understand skin tolerability and safety. Factors influencing product trust, including formulation composition (herbal vs. conventional analgesics) and institutional credibility were also recorded alongside participant intent to reuse or recommend the patch.

Results

The prepared herbal patches were formulated successfully having transparent, and easy peelable properties. The study possesses desirable physical and mechanical characteristics, indicating that the selected components were suitable for transdermal formulation and method of preparation. The patches appeared smooth, flexible and transparent, indicating uniform distribution of the adhesive and herbal constituents within the matrix. Transparent patches indicate the absence of excipients or drug crystallization, good compatibility between drug and excipients.**[24]**Additionally, the drug adhesive patches were successfully formulated and showed proper adhesion with the backing layer, which is a critical parameter for transdermal drug delivery. Proper adhesion ensures that the patch remains intact during application, thereby obtaining consistent drug delivery through the skin. The compatibility between adhesive layer and backing membrane indicates that the selected materials were appropriate and effective in transdermal patches. The photographic image of the prepared herbal patch is shown in **Figure 2.**

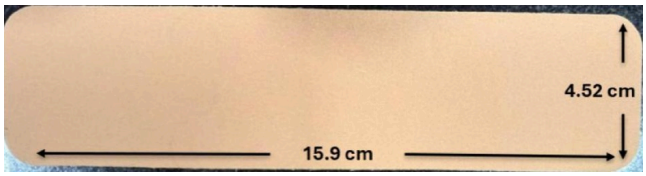


Fig. 2: Prepared Ayurvedic herbal transdermal patch used for menstrual pain management. The developed patch measured approximately 15.9 cm × 4.52 cm and exhibited a smooth, transparent, and flexible appearance suitable for topical application.

Evaluation of Herbal patches

The folding endurance, thickness, weight variation, & percentage elongation of prepared herbal patches shows optimum consideration. The prepared herbal patches exhibit folding endurance of 100.33 & 45% elongation, signifying excellent mechanical strength and elasticity. The variation in thickness and weight of herbal patch, at 0.63mm and 3.31g respectively, suggests uniform distribution of adhesive and herbal drug mixtures throughout patch. The results of evaluated parameters were given in Table no. 2.

Table 2: Characterization results of the formulated herbal transdermal patch.

Name of the Parameters	Results
Folding endurance	100.33 ± 1.52
Thickness	0.63 ± 0.90 mm
Weight variation	3.31 ± 0.20 g
Percentage elongation	45%

Ex-vivo Skin Permeation studies

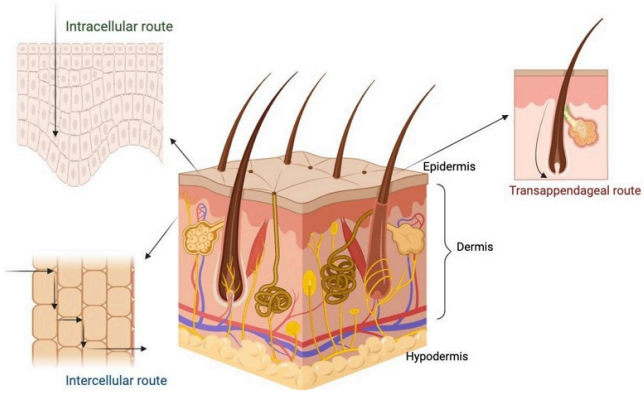


Fig. 3: Schematic representation of transdermal drug permeation pathways through skin. Illustration depicts intercellular, transcellular, and appendageal routes involved in transport of active herbal constituents across stratum corneum and into systemic circulation. The image was made using BioRender illustration tool (*BioRender, Toronto, ON, Canada*).

Transdermal permeation profiles were evaluated ex-vivo utilizing freshly prepared porcine skin membranes. The % Cumulative drug release of the developed herbal patch was evaluated over a period of 12 hours to ascertain the quantity of the drug permeated through the skin as shown in **Figure 4.** The formulation exhibited an initial release of 7.69µg/ml at one hour, followed by a sustained release pattern, reaching 2.26µg/ml at 12 hours. A biphasic release profile was observed, wherein an initial burst effect was followed by prolonged delivery over 12 hours. The study also revealed a time dependent increase in cumulative drug permeation across the membrane. It was found that the total amount of medication absorbed by each patch was 13687 µg/patch at 12 hours, indicating effective permeation of the drug through the membrane. The maximum steady-state flux (J) was found to be 465.51 µg/cm2/hr, this indicates the permeation of drug is better with dibutyl phthalate used as a permeation enhancer.

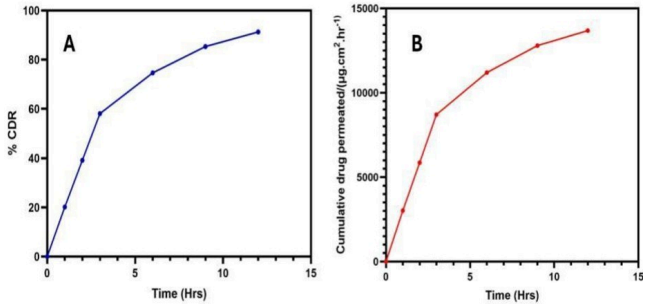


Fig. 4: Ex-vivo permeation profile of the optimized herbal transdermal patch showing (A) percentage cumulative drug release and (B) cumulative drug permeation through porcine skin over 12 hours. Graphs were generated using Origin software (*OriginLab Corporation, Northampton, MA, USA*)

In-vivo translational validation (patient feedback)

This laboratory-observed release kinetics maps directly to real-world patient outcomes evaluated in a pilot user cohort (n = 27). The immediate release phase successfully addressed the acute clinical need for rapid menstrual pain management, yielding an overall therapeutic response rate of 96% (n = 26). Within this responding cohort, 63% (n = 17) of human participants reported noticeable pain relief within first 15 minutes of patch application. An additional 22% (n = 6) experienced relief within 1530 minutes, marking a combined early clinical response rate of 85% within 30 minutes post-application.

Only a single participant (4%) reported zero perceptible pain relief. Notably, 96% of users (n = 26) reported that the patch remained perfectly in place during their daily routine activities throughout its application duration. The patient-reported analgesic efficacy yielded a mean clinical score of 4.2 on a 5-point Likert scale. Within this cohort, 74% of respondents (n = 20) assigned high-efficacy ratings of either four or five, with 52% (n = 14) stating that the patch completely eliminated or near-completely managed their cramps (assigning the maximum rating of five).

Stability Studies

1. Accelerated Stability Results (40°C ± 2°C / 75% RH ± 5%)

The accelerated stability data for the Ayurvedic herbal transdermal patch over a three-month period. All tested parameters remained within prespecified acceptance criteria throughout the study duration.

The physical description of the patch remained unchanged at the three-month accelerated time point, indicating no visible signs of phase separation, discolouration, or structural degradation. High Pressure Liquid Chromatography confirmed the presence of both active constituents at all time points, with peak retention times consistent with respective reference standard solutions. Patch dimensions remained within specification at both T0 and T3M under accelerated conditions. Adhesive strength declined marginally from 137.0 gm (T0) to 130.5 gm (T3M), both values exceeding the minimum specification of 42 g/width 12 mm. The Boswellia serrata assay was 106.39% at T0, declining to 105.0% at 3 months. Menthol content was 100.0% at T0, decreasing to 99.25% at T3M, remaining well within specification.

2. Long-Term Stability Results (30°C ± 2°C / 65% RH ± 5%)

Under long-term conditions, the product description, identification, and dimensional parameters showed no significant change. Adhesive strength decreased from 137.0 gm (T0) to 128.5 gm at T3M, remaining well above the minimum specification. The Boswellia serrata assay showed a minimal decline from 106.39% to 104.98%, and menthol content remained stable at 99.0% compared to the initial value of 100.0%.

Discussion

The successful development of herbal transdermal patch demonstrates suitability of the selected herbal constituents, adhesive matrix & permeation enhancer for transdermal drug delivery. The prepared patches exhibited desirable physicochemical and mechanical characteristics, including uniform thickness, weight variation, folding endurance, and percentage elongation. These properties indicate adequate flexibility, structural integrity, and uniform distribution of herbal drug mixture within adhesive matrix, all of which are essential requirements for effective transdermal application. Transparent & smooth appearance of patches further suggests good compatibility among formulation components and absence of visible drug crystallization within matrix.

The ex-vivo permeation study revealed a biphasic release profile characterized by an initial burst release followed by sustained drug delivery over a period of 12 hours. This phenomenon is mediated by the localization of the active compounds near the matrix boundary, which promotes rapid initial diffusion. This immediate release majorly helps in the management of menstrual pain. The release profile indicates sustained drug release behaviour indicates adhesive matrix effectively regulated drug diffusion through combined effect of diffusion controlled and matrix relaxation mechanism.**[25]** The cumulative drug permeation and flux values obtained in this study further demonstrate the efficiency of the developed formulation. Previous studies on NSAID-based and herbal transdermal patches have demonstrated equivalent cumulative permeation values over similar durations, depending on formulation composition and permeation enhancers.**[26]**

The enhanced permeation observed in the present study indicates efficient drug-matrix interaction and effective transport of the active constituents across the skin membrane.. The maximum steady-state flux (J) was found to be 465.51 µg/cm²/hr, suggesting enhanced permeation of the drug through the membrane, potentially due to the presence of dibutyl phthalate as a permeation enhancer. For instance, studies involved with ion-pair or enhancer-based systems have reported flux values ranging from moderate to high values of µg/cm²/hr, depending on drug properties and enhancement strategies.**[27]**

The markedly higher flux observed in the present study suggests superior permeation characteristics of the developed formulation and highlights its potential to achieve therapeutically relevant drug concentrations through transdermal administration.

The findings also provide a theoretical basis for combining multiple herbal constituents with complementary pharmacological activities within a single transdermal platform. While *Peppermint Satva* primarily contributes rapid analgesic and cooling effects through menthol-mediated modulation of pain perception, *Shallaki Guggul* and *Haridra Satva* provide sustained anti-inflammatory activity through the suppression of inflammatory mediators.**[28]***Taila Parni* acts as a natural permeation enhancer, facilitating transdermal transport of the active constituents, while Cinnamon Oil contributes additional antispasmodic and anti-inflammatory effects. The combination of these herbal ingredients may therefore provide a synergistic therapeutic effect by simultaneously targeting uterine smooth muscle spasm, inflammation, prostaglandin-mediated pain pathways, and pain perception.**[29]**

From a practical perspective, the developed patch offers several advantages over conventional oral analgesics. The transdermal route bypasses the gastrointestinal tract, thereby reducing the likelihood of gastrointestinal irritation commonly associated with prolonged NSAID use. In addition, the sustained-release nature of the formulation may reduce the need for repeated dosing and improve patient compliance. The non-invasive mode of administration, ease of application, prolonged wearability, and high user acceptance observed in the present study further support its potential suitability for routine menstrual pain management.

The stability studies demonstrated that the formulation remained physically and chemically stable under both accelerated and long-term storage conditions. The minimal changes observed in adhesive strength and active constituent content over the study period indicate that the formulation possesses adequate stability for storage, transportation, and potential future commercialization.

Despite the encouraging findings, certain limitations should be acknowledged. The user acceptability assessment involved a relatively small cohort and relied primarily on self-reported outcomes.

Furthermore, ex-vivo permeation studies using porcine skin may not completely replicate human skin physiology.**[30]**Therefore, larger randomized clinical studies are required to establish clinical efficacy, dose-response relationships, pharmacokinetic behaviour, and long-term safety in diverse populations.

Overall, the results suggest that the developed herbal transdermal patch successfully combines rapid onset of action with sustained drug delivery, providing a promising non-invasive approach for the management of menstrual pain. The synergistic action of the herbal constituents, together with efficient transdermal permeation, formulation stability, and favourable user acceptance, highlights the potential of this formulation as an alternative to conventional oral therapies for menstrual pain management.

Conclusion

The present study successfully developed an herbal-based transdermal patch for the management of menstrual pain using a combination of natural bioactive ingredients with complementary pharmacological actions. The formulation demonstrated excellent physicochemical and mechanical properties, indicating suitability for transdermal application. The incorporation of acrylic pressure sensitive adhesive and permeation enhancers contributed to uniform drug distribution, structural integrity, and effective skin permeation. Ex-vivo permeation studies confirmed a pattern of biphasic release with an initial quick drug release, then prolonged delivery over 12 hours, which is particularly beneficial for immediate and prolonged pain relief during menstrual pain. High cumulative drug release and enhanced flux values indicate superior permeation characteristics compared to conventional formulations. Importantly, patient feedback strongly validated the laboratory findings. The patch showed a fast response rate, with nearly 80% of participants feeling pain relief within 30 minutes, excellent staying power during daily activities, and good skin safety with no side effects. High user satisfaction, trust in product & strong willingness to recommend it demonstrate its seamless compatibility for managing menstrual pain. Finally, helpful user feedback about adjusting the design and scent gives us a clear roadmap for improving patch as we move toward larger scale production.

The synergistic action of herbal components provided anti-inflammatory, antispasmodic, and analgesic effects while effectively reducing the risk of systemic and gastrointestinal adverse effects of oral NSAIDs.

Additionally, the transdermal route improved patient compliance by offering a non-invasive, controlled drug delivery system. In conclusion, the developed herbal transdermal patch represents a promising, safe, and effective alternative to conventional therapies for menstrual pain.

Further knowledge regarding in-vivo and clinical studies with a much larger cohort are recommended to validate its therapeutic efficacy and long-term safety.

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